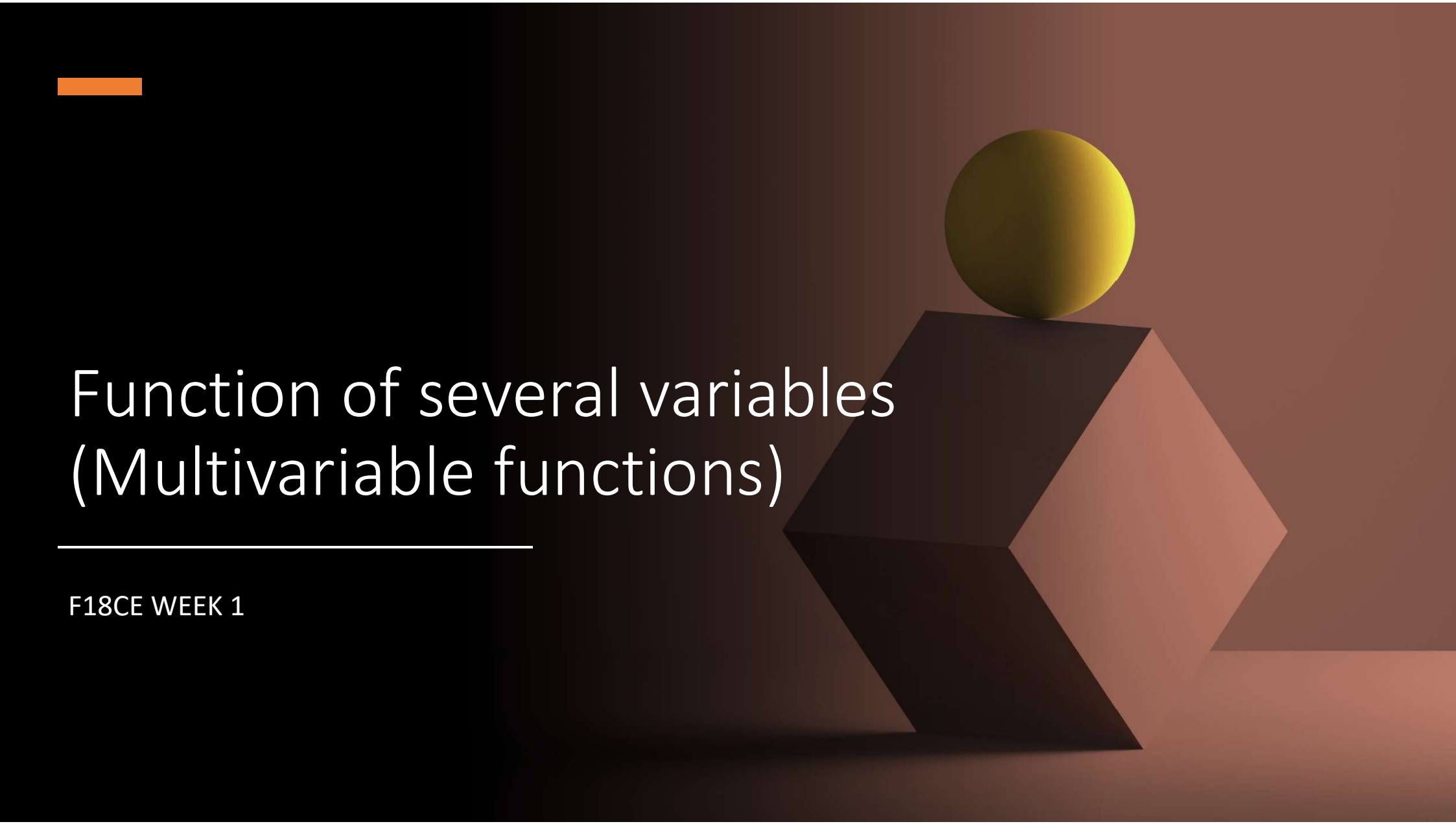


Function of several variables (Multivariable functions)

F18CE WEEK 1

Introduction – Multivariable function

Let $f(x_1, x_2, \ldots, x_n) : \mathbb{R}^n \to \mathbb{R}^m$ be a multivariable function.

$$f : \mathbb{R}^2 \to \mathbb{R} \quad f(x, y) = x^2 + \sin y$$

For example, $g : \mathbb{R}^2 \to \mathbb{R} \quad g(x, y) = x^2 y^3$

$$h : \mathbb{R}^n \to \mathbb{R} \quad h(x_1, \ldots, x_n) = x_1 + \ldots + x_n$$

Remark: In this course, we focus more on the case when $m = 1$.

Let $f(x) = x^{\mu}$. We study the domain of f for several cases of μ.

Domain of f is highlighted in yellow:

	$\mu > 0$	*Example*	$\mu < 0$	*Example*
$\mu \in \mathbb{Z}$	$\mathbb{R}$	$f(x) = x^2$	$\mathbb{R}\backslash\{0\}$	$f(x) = x^{-3}$
$\mu = \frac{m}{2n+1} \in \mathbb{Q}\ (m, n \in \mathbb{Z})^*$	$\mathbb{R}$	$f(x) = x^{2/3}$	$\mathbb{R}\backslash\{0\}$	$f(x) = x^{-1/3}$
$\mu = \frac{m}{2n} \in \mathbb{Q}\ (m, n \in \mathbb{Z})^*$	$\mathbb{R}\backslash\mathbb{R}^-$	$f(x) = x^{1/2}$	$\mathbb{R}^+$	$f(x) = x^{-1/2}$
$\mu \in \mathbb{R}\backslash\mathbb{Q}$	$\mathbb{R}\backslash\mathbb{R}^-$	$f(x) = x^{\sqrt{2}}$	$\mathbb{R}^+$	$f(x) = x^{-\pi}$

*The fraction is in the simplest form

Or,

μ	0	$\mathbb{Z}\backslash\{0\}$		$\mathbb{Q}\backslash\{0\}$ (μ is written in the simplest form)				$\mathbb{R}\backslash\mathbb{Q}$	
		$\mathbb{Z}^+$	$\mathbb{Z}^-$	$\mu = \frac{m}{2n} > 0$	$\mu = \frac{m}{2n} < 0$	$\mu = \frac{m}{2n+1} > 0$	$\mu = \frac{m}{2n+1} < 0$	$\mu > 0$	$\mu < 0$
Domain	$\mathbb{R}\backslash\{0\}$	$\mathbb{R}$	$\mathbb{R}\backslash\{0\}$	$\mathbb{R}\backslash\mathbb{R}^-$	$\mathbb{R}^+$	$\mathbb{R}$	$\mathbb{R}\backslash\{0\}$	$\mathbb{R}\backslash\mathbb{R}^-$	$\mathbb{R}^+$

Remark. 1. When $\mu = \frac{m}{n} \in \mathbb{Q}$, $y = f(x) = x^{\mu}$ is defined as $\exists y \in \mathbb{R} \ni y^n = x^m$.

2. When μ is irrational, $f(x) = x^{\mu}$ is defined as $f(x) = x^{\mu} = e^{\mu \ln x}$, so $x > 0$.

3. When $\mu = 0$, $f(x) = x^{\mu} = x^0 = 1$ for all $x \in \mathbb{R} \setminus \{0\}$.

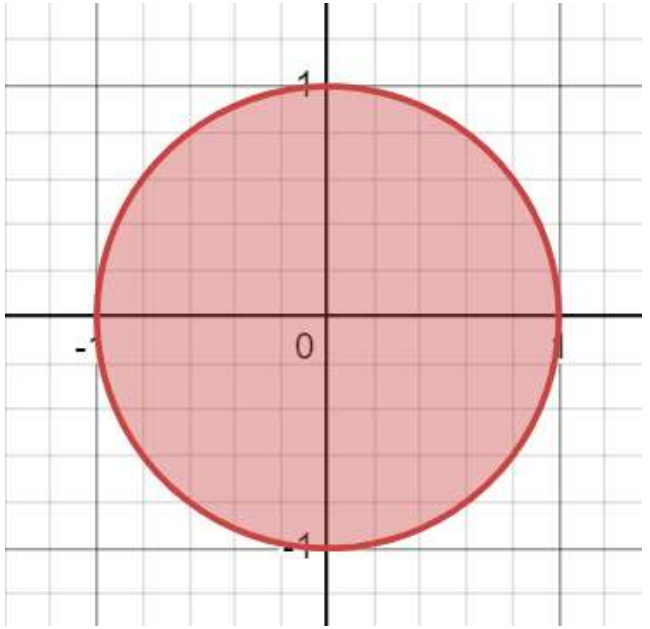
4. 0^0 may be interpreted differently; it is considered undefined or indeterminate form in analysis; but defined as 1 in combinatorics or abstract algebra.

5. When $x = 0$, $0^{\mu} = 0$ for all reals $\mu > 0$. So, the case when μ is positive irrational, domain of $f(x) = x^{\mu}$ is $\mathbb{R} \setminus \mathbb{R}^-$ (i.e., include 0).

Example 1.1

Write the domain for $f(x,y) = \sqrt{1 - x^2 - y^2}$.

Solution. $D_f = \left\{(x,y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\right\}$.



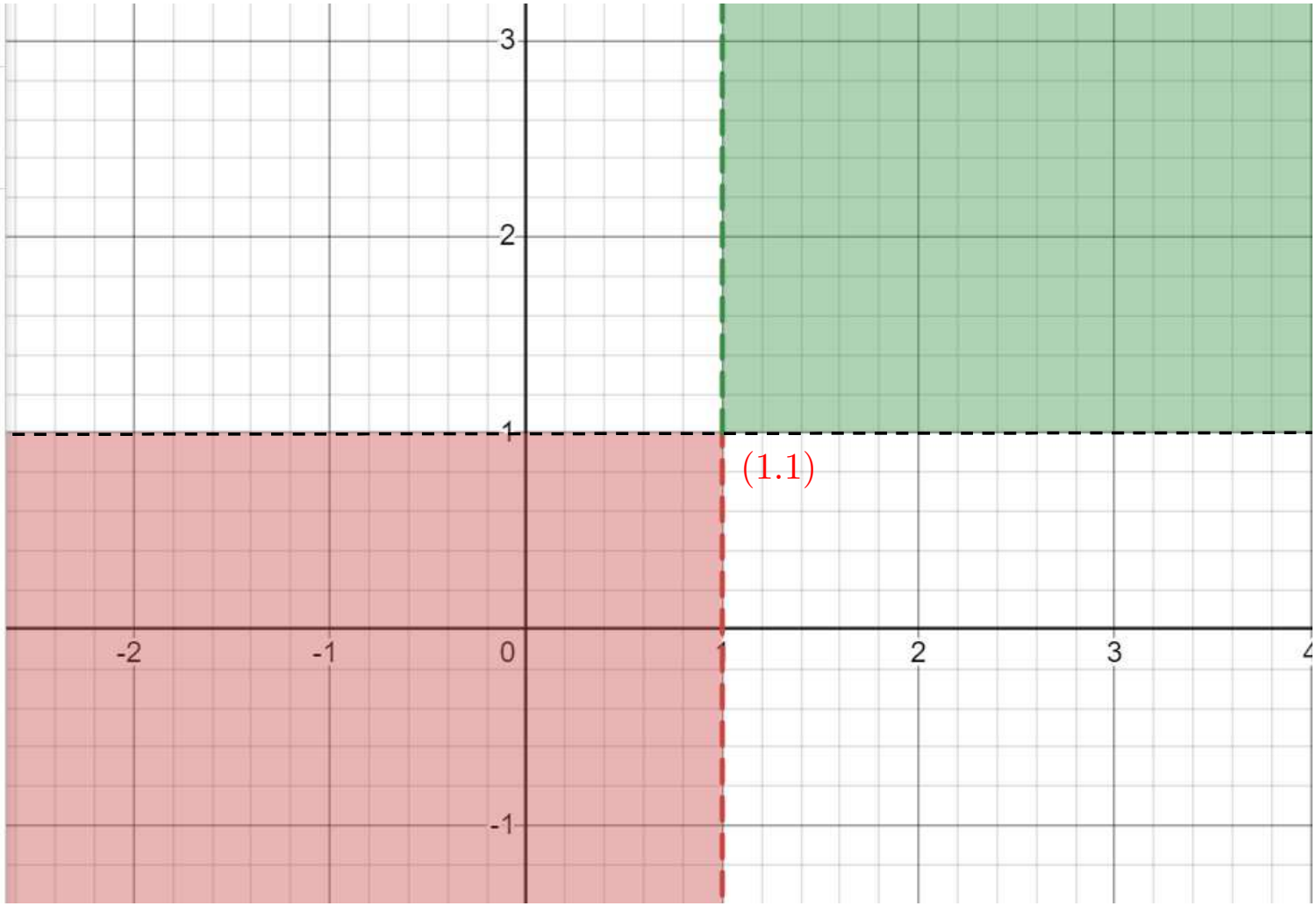
Example 1.2

Write the domain for $f(x,y) = \ln\Big[(1-x)(1-y)\Big]$.

Solution. Note that $(1-x)(1-y) > 0$ so that the natural log (ln) is defined. So, we have either

$$1-x>0 \;\;\&\;\; 1-y>0\,;\text{ or } 1-x<0 \;\;\&\;\; 1-y<0\,.$$

Thus, $D_f = \Big\{(x,y)\in\mathbb{R}^2 : x,y<1 \text{ or } x,y>1\Big\}$.


3
2
1
-2
-1
0
1
2
3
-1
(1.1)

Example 1.3

Let $f : \mathbb{R} \to \mathbb{R}$ be a function and define function g to be

$$g(x, y) = \sqrt{y} + f(\sqrt{x} - 1)\,.$$

Write the domain of g.

Solution. $D_g = \left\{(x, y) \in \mathbb{R}^2 : x, y \geq 0\right\}.$

Example 1.4

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined as $f(x, y) = \frac{x^4}{y^2} - 2xy$.

Define a new function $g : \mathbb{R} \to \mathbb{R}$ by $g(x) = f\left(x, \frac{x}{2}\right)$.

Write the explicit formula for $g(x)$.

Solution. $g(x) = f\left(x, \frac{x}{2}\right) = \frac{x^4}{\left(\frac{x^2}{4}\right)} - 2x\left(\frac{x}{2}\right) = 4x^2 - x^2 = 3x^2$.

Example 1.5

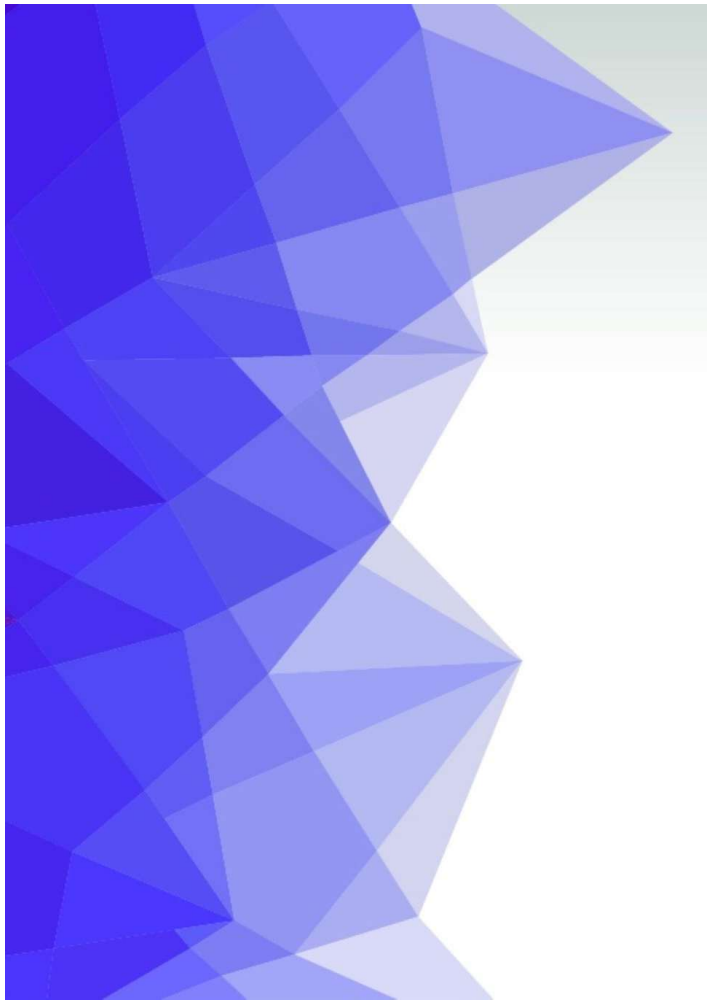
Let $f : \mathbb{R} \to \mathbb{R}$ be a function and

$g : \mathbb{R}^2 \to \mathbb{R}$ be a function defined by

defined as $g(x, y) = \sqrt{y} + f\left(\sqrt{x} - 1\right)$.

Write the explicit formula for f if $g(x, 1) = x$.

$$x = g(x,1) = \sqrt{1} + f\left(\sqrt{x}-1\right) = 1 + f\left(\sqrt{x}-1\right)$$
$$\Rightarrow f\left(\sqrt{x}-1\right) = x-1 = \left(\sqrt{x}-1\right)\left(\sqrt{x}+1\right) = \left(\sqrt{x}-1\right)\left(\sqrt{x}-1+2\right).$$
$$\Rightarrow f(w) = w(w+2)$$



Limits of functions of two variables

Recall definition of limit for single variable function

Let $f : D \to \mathbb{R}$ be a function and $c \in \overline{D}$.

Then f converges to $L \in \mathbb{R}$ as x tends to c,

written as $\lim_{x \to c} f(x) = L$ if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \text{ such that } \left| f(x) - L \right| < \varepsilon$$

$$\text{whenever } 0 < \left| x - c \right| < \delta \text{ and } x \in D\,.$$

Definition 1.6: Limits of functions of two variables

Let $f : D \to \mathbb{R}$ where $D \subseteq \mathbb{R}^2$.

Then f converges to $L \in \mathbb{R}$ at $\Big(a, b\Big) \in \mathbb{R}^2$ (may not be in D),

written as $\lim\limits_{(x,y)\to(a,b)} f(x,y) = L$ if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \text{ such that } \Big|f(x,y) - L\Big| < \varepsilon$$

$$\text{whenever } 0 < \Big|x - a\Big| < \delta \,,\ 0 < \Big|y - b\Big| < \delta \text{ and } (x,y) \in D\,.$$

Equivalence definition (using norm)

Equivalently, by using the Euclidean norm in the plane,

which means that we compute the distance from $\Big(x, y\Big)$ to $\Big(a, b\Big)$ as

$$\operatorname{dist}\Big((x,y),(a,b)\Big) = \sqrt{(x-a)^2 + (y-b)^2} := \Big\|(x,y) - (a,b)\Big\|$$

(which generalises to any dimension) this definition can be stated as

Definition 1.7 (equivalent to Definition 1.6)

Let $f : D \to \mathbb{R}$ where $D \subseteq \mathbb{R}^2$.

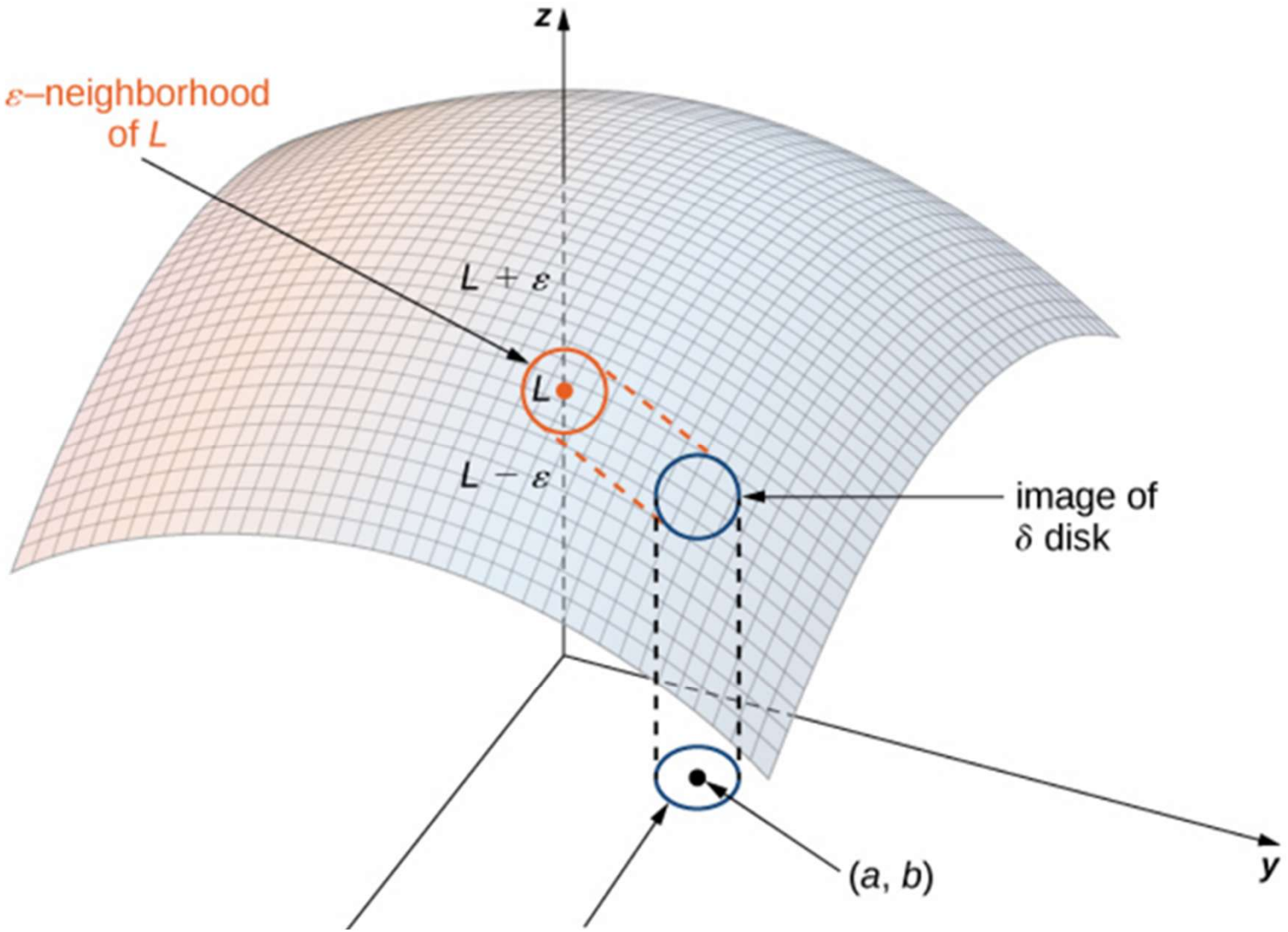
Then f converges to $L \in \mathbb{R}$ at $\left(a,b\right) \in \mathbb{R}^2$ (may not be in D),

written as $\lim\limits_{(x,y)\to(a,b)} f(x,y) = L$ if

$\forall \varepsilon > 0 \ \exists \delta > 0$ such that $\left|f(x,y) - L\right| < \varepsilon$

whenever $0 < \left\|(x,y) - (a,b)\right\| < \delta$

(i.e., $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$), and $(x,y) \in D$.


z
ε–neighborhood
of L
L + ε
L
L − ε
image of
δ disk
(a, b)
y

Theorem 1.9

Let $f, g : D \to \mathbb{R}, D \subseteq \mathbb{R}^2$ such that

$$\lim_{(x,y)\to(a,b)} f(x,y) = L \text{ and } \lim_{(x,y)\to(a,b)} g(x,y) = M\,.$$

Then for any $\alpha, \beta \in \mathbb{R}$ (a) $\lim_{(x,y)\to(a,b)} (\alpha f + \beta g) = \alpha L + \beta M$

(b) $\lim_{(x,y)\to(a,b)} fg = LM$ (c) $\lim_{(x,y)\to(a,b)} \frac{f}{g} = \frac{L}{M}, M \neq 0$

(d) Suppose $h : \mathbb{R}^2 \to \mathbb{R}^2$ such that $\lim_{(x,y)\to(c,d)} h(x,y) = (a,b)\,.$

Then $\lim_{(x,y)\to(c,d)} f \circ h = L\,.$

Example 1.16

$$\lim_{(x,y)\to(2,3)} \frac{x+\sin y}{x^2y^2+1} = \frac{2+\sin 3}{4(9)+1} = \frac{2+\sin 3}{37} \quad (\sin 3 \text{ is measured in radian})$$

Theorem 1.8

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function with domain D.

The following statements are equivalent

(a) $\lim\limits_{(x,y)\to(a,b)} f(x,y) = L$

(b) For any pair of sequences $\left(x_n, y_n\right) \in D \setminus \left\{\left(a,b\right)\right\} \forall n \in \mathbb{N}$ such that

$\lim\left(x_n\right) = a$, $\lim\left(y_n\right) = b$ (i.e., $\left(x_n, y_n\right) \to \left(a,b\right)$ as $n \to \infty$), we have

$$\lim_{n\to\infty} f\left(x_n, y_n\right) = L\,.$$

How to apply Theorem 1.8?

If there exist two pairs of sequences

$\left(x_n, y_n\right), \left(\tilde{x}_n, \tilde{y}_n\right)$ with $\left(x_n, y_n\right) \to \left(a, b\right)$

and $\left(\tilde{x}_n, \tilde{y}_n\right) \to \left(a, b\right)$ as $n \to \infty$;

but $\lim\limits_{n\to\infty} f\left(x_n, y_n\right) \neq \lim\limits_{n\to\infty} f\left(\tilde{x}_n, \tilde{y}_n\right)$. Then

$$\lim_{(x,y)\to(a,b)} f(x,y) \text{ does not exist.}$$

Example 1.10

Let $g(x,y) = \frac{x^2-y^2}{x^2+y^2}$ defined for $(x,y) \in \mathbb{R}^2 \setminus \{(0,0)\}$.

Show that $\lim\limits_{(x,y)\to(0,0)} g(x,y)$ does not exist.

Solution. Note that $\left(\frac{1}{n},\frac{1}{n}\right) \to (0,0)$ and $\left(\frac{1}{n},0\right) \to (0,0)$ as $n \to \infty$.

Now, $g\left(\frac{1}{n},\frac{1}{n}\right) = \frac{1/n^2-1/n^2}{1/n^2+1/n^2} \to 0$ as $n \to \infty$ and $g\left(\frac{1}{n},0\right) = \frac{1/n^2-0}{1/n^2+0} \to 1$ as $n \to \infty$.

Since $\lim\limits_{n\to\infty} g\left(\frac{1}{n},\frac{1}{n}\right) \neq \lim\limits_{n\to\infty} g\left(\frac{1}{n},0\right)$, it follows that $\lim\limits_{(x,y)\to(0,0)} g(x,y)$ does not exist.

Example 1.17

Determine if $f(x,y) = x^y, x, y \in \mathbb{R}^+$ can be made continuous at (0, 0).

Solution. Note that f is undefined at (0, 0).

Pick $\left(x_n, y_n\right) = \left(0, \frac{1}{n}\right) \to \left(0,0\right)$, then $\lim\limits_{n\to\infty} f\left(x_n, y_n\right) = \lim\limits_{n\to\infty} 0^{1/n} = \lim\limits_{n\to\infty} 0 = 0$.

Pick $\left(x_n, y_n\right) = \left(\frac{1}{n}, 0\right) \to \left(0,0\right)$, then $\lim\limits_{n\to\infty} f\left(x_n, y_n\right) = \lim\limits_{n\to\infty} \left(\frac{1}{n}\right)^0 = \lim\limits_{n\to\infty} 1 = 1$.

Even if we assign $f(0,0) = 0$ or 1, the function is still not continuous,

because $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ does not exist.

Alternative method 1: Extra example (a)

(a) Evaluate $\lim\limits_{(x,y)\to(0,0)} \frac{x^2y}{x^2+y^2}$.

Solution. Let $x = r\cos\theta$, $y = r\sin\theta$. Then

$$\frac{x^2y}{x^2+y^2} = \frac{r^3\cos^2\theta\sin\theta}{r^2\cos^2\theta+r^2\sin^2\theta} = r(1-\sin^2\theta)\sin\theta = r\cos^2\theta\sin\theta.$$

Since $r^2 = x^2 + y^2 \to 0$ as $(x,y)\to(0,0)$. Thus

$$\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^2+y^2} = \lim_{r\to 0^+} r\cos^2\theta\sin\theta = 0$$

because $\forall\theta\in\mathbb{R}$, $\left|\cos^2\theta\sin\theta\right| \le 1$. To be exact: $\left|\cos^2\theta\sin\theta\right| \le \frac{2\sqrt{3}}{9}$

Remark: Here we apply the Squeeze theorem.

Alternative method 2: Extra example (b)

$$\text{Let } f(x,y) = \begin{cases} \frac{(2x-y)^2}{x-y}, & x \neq y \\ 0, & x = y \end{cases}.$$

$$\text{Show that } \lim_{(x,y)\to(0,0)} f(x,y) \text{ does not exist.}$$

Consider the pathway $y = 2x$:

We have $\frac{(2x-y)^2}{x-y} = \frac{(2x-2x)^2}{x-2x} = 0$ and

thus $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0$

Consider the pathway $y = x + x^2$:

$\frac{(2x-y)^2}{x-y} = \frac{(2x-x-x^2)^2}{x-x-x^2} = -\frac{x^2(1-x)^2}{x^2} = -(1-x)^2$

so $\lim\limits_{(x,y)\to(0,0)} f(x,y) = -1 \neq 0$.

Hence, the limit does not exist.

This approach does not work!!

Suppose $x = r\cos\theta\,,\ y = r\sin\theta\,.$ Then

$$f(r\cos\theta, r\sin\theta) = \begin{cases} r\,\frac{(2\cos\theta - \sin\theta)^2}{\cos\theta - \sin\theta}, & \cos\theta \neq \sin\theta \\ 0, & \cos\theta = \sin\theta \end{cases}.$$

Note that $\cos\theta = \sin\theta$ iff $\theta = \frac{\pi}{4} + n\pi, n \in \mathbb{Z}\,.$

No conclusion because we cannot guarantee the existence of $\lim\limits_{(x,y)\to(0,0)} f(x,y)\,.$

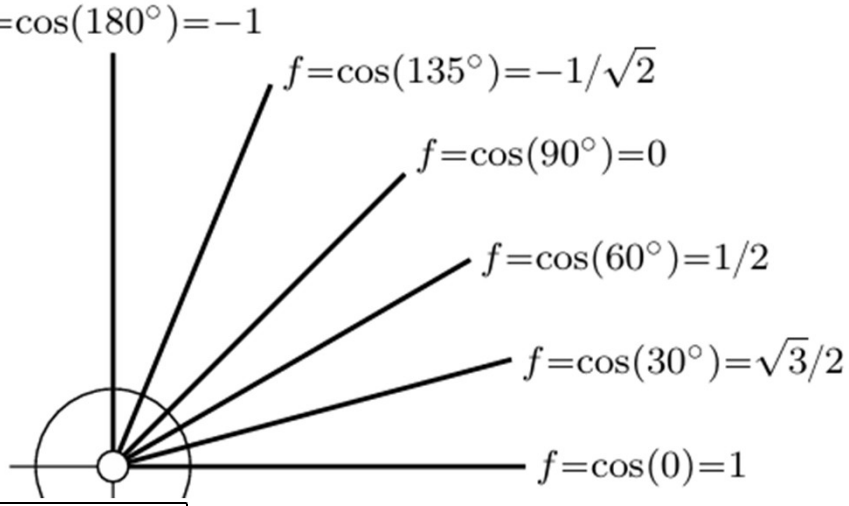
Revisit Example 1.10

Show that $\lim\limits_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$ does not exist.

Solution. Let $x = r\cos\theta$, $y = r\sin\theta$. Then

$$\frac{x^2-y^2}{x^2+y^2} = \frac{r^2(\cos^2\theta-\sin^2\theta)}{r^2(\cos^2\theta+\sin^2\theta)} = \cos 2\theta\,.$$

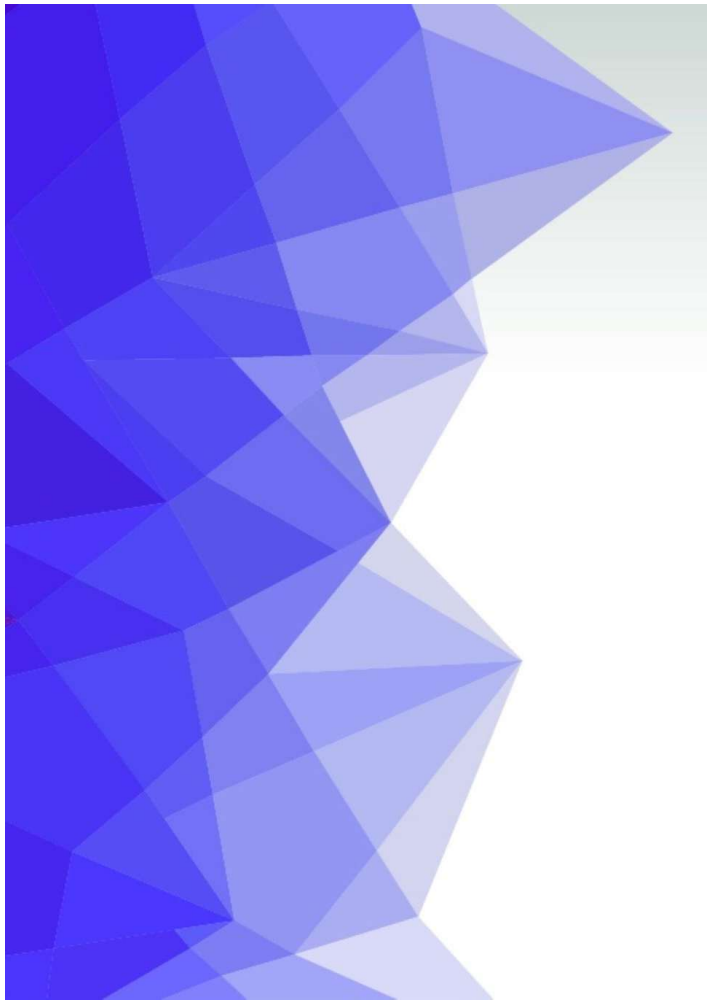
Note that $\lim\limits_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2}$ does not exist because:



$(x,\ y)$ approaches $(0,\ 0)$ along the line	$\theta = 0$	$\theta = \pi/2$
$\frac{x^2-y^2}{x^2+y^2} = \cos 2\theta$	1	-1 $(\neq 1)$

Summary

1. Use Theorem 1.8. Find two pairs of sequences $\left(x_n, y_n\right), \left(\tilde{x}_n, \tilde{y}_n\right) \to \left(a, b\right)$ as $n \to \infty$; such that $\lim_{n\to\infty} f\left(x_n, y_n\right) \neq \lim_{n\to\infty} f\left(\tilde{x}_n, \tilde{y}_n\right)$. Then $\lim_{(x,y)\to(a,b)} f(x,y)$ does not exist.
2. If there exist two pathways $y = g(x)$ and $y = h(x)$ to $\left(a, b\right)$ such that $\lim_{(x,y)\to(a,b)} f\left(g(x), y\right) \neq \lim_{(x,y)\to(a,b)} f\left(h(x), y\right)$. Then $\lim_{(x,y)\to(a,b)} f(x,y)$ does not exist.
3. If $f(x,y) = f(r\cos\theta, r\sin\theta)$ can be expressed as $rg(\theta)$ where $g(\theta)$ is bounded, then $\lim_{(x,y)\to(0,0)} f(x,y) = 0$ (only applicable to $(x,y) \to (0,0)$).
4. Let $f(x,y) = f(r\cos\theta, r\sin\theta) = g(\theta)$. If there exist two pathways such that $g(\theta_1) \neq g(\theta_2)$, then $\lim_{(x,y)\to(a,b)} f(x,y)$ does not exist.



Continuity in multivariable function

Definition 1.11. Continuity at a point for a function of two variables

A function of two variables $f(x,y)$ with domain $D_f \subset \mathbb{R}^2$ is called continuous at a point $\left(a,b\right) \in D_f$ if $\lim\limits_{(x,y)\to(a,b)} f(x,y) = f(a,b)$. A function f is said to be continuous (anywhere in its domain) if f is continuous at any point in D_f.

In view of Definition 1.11

f is not continuous at (a,b) if $\lim_{(x,y)\to(a,b)} f(x,y)$ does not exist.

Remark: Of course it is also true to say that

f is not continuous at (a,b) if

$f(a,b)$ is not defined or $\lim_{(x,y)\to(a,b)} f(x,y) \neq f(a,b)$.

Both these two cases are trivial.

Example 1.12

Consider $f(x,y) = \begin{cases} \frac{xy}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$ defined on entire $\mathbb{R}^2$.

Show that $f(x,y)$ is not continuous at (0, 0).

Solution. Note that $\left(\frac{1}{n},\frac{1}{n}\right),\left(\frac{1}{n},0\right) \to \left(0,0\right)$ as $n \to \infty$.

$$\lim_{n\to\infty} f\left(\tfrac{1}{n},\tfrac{1}{n}\right) = \tfrac{1}{2} \neq 0 = \lim_{n\to\infty} f\left(\tfrac{1}{n},0\right).$$

So, $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0$ does not exist and so it is not continuous at (0, 0).

Example 1.13

Consider a function of two variables

$$f(x,y) = \begin{cases} \frac{x^2 y}{x^2+y^2}, (x,y) \neq (0,0) \\ 0, \quad (x,y) = (0,0) \end{cases}$$

defined on the entire $\mathbb{R}^2$.

Show (intuitionally) that this function is continuous at (0,0).

Solution. Let $x = r\cos\theta, y = r\sin\theta$. Then

$$\frac{x^2y}{x^2+y^2} = \frac{r^3\cos^2\theta\sin\theta}{r^2(\cos^2\theta+\sin^2\theta)} = r\cos^2\theta\sin\theta\ .$$

So, $\lim\limits_{(x,y)\to(0,0)} \frac{x^2y}{x^2+y^2} = \lim\limits_{r\to 0^+} r\cos^2\theta\sin\theta = 0 = f(0,0)$

as $\left|\cos^2\theta\sin\theta\right| \leq 1\ \forall\theta\in\mathbb{R}$ (refer to Extra example (a)).

This implies that the continuity of the function at the origin.

Note that this is not a fully rigorous way to prove it.

Theorem 1.15

Let $f, g : \mathbb{R}^n \to \mathbb{R}$ be two multivariable functions that are each continuous at $\mathbf{a} = \left(a_1, a_2, \ldots, a_n\right) \in \mathbb{R}^n$. Then

(i) $\alpha f(\mathbf{x}) + \beta g(\mathbf{x})$ is continuous at $\mathbf{a}$, where $\alpha, \beta \in \mathbb{R}$ are scalar constants.

(ii) $f(\mathbf{x}) \cdot g(\mathbf{x})$ is continuous at $\mathbf{a}$.

(iii) $\dfrac{f(\mathbf{x})}{g(\mathbf{x})}$ is continuous at $\mathbf{a}$, provided that $g(\mathbf{a}) \neq \mathbf{0}$.

Let $h : \mathbb{R}^n \to \mathbb{R}$ and consider $h \circ f(\mathbf{x}) = h\Big(f(\mathbf{x})\Big)$. Then

(iv) $f(\mathbf{x})$ is continuous at $\mathbf{a}$ implies that $h(\mathbf{x})$ is continuous at $\mathbf{a}$.

Example 1.14

The function $h(x,y) = \begin{cases} \frac{x^2+y^2}{x^2-y^2}, & x^2 \neq y^2 \\ 1, & x^2 = y^2 \end{cases}$ is defined on the entire $\mathbb{R}^2$.

Show that the function h has a discontinuity along a pair of lines $\pm x = y$,

i.e., h is discontinuous at each point on lines $y = x$ and $y = -x$.

Both x^2+y^2 and x^2-y^2 are continuous on $\mathbb{R}^2$ (Theorem 1.15(iii))

and so $h(x,y)=\frac{x^2+y^2}{x^2-y^2}$ is continuous on $\mathbb{R}^2$, except for possibly $x^2=y^2$.

We claim that h is not continuous on $D=\left\{(k,\pm k):k\in\mathbb{R}\right\}$. This is because $\left(\frac{1}{n}+k,k\right),\left(k,\frac{1}{n}+k\right)\to\left(k,k\right)$ as $n\to\infty$, and applying L'hopital's rule:

$$\lim_{n\to\infty}h\left(\tfrac{1}{n}+k,k\right)=\lim_{n\to\infty}\tfrac{1/n^2+2k/n+2k^2}{1/n^2+2k/n}=\lim_{n\to\infty}\tfrac{1+2kn+2n^2k^2}{1+2kn}=\lim_{n\to\infty}(1+2kn)=\begin{cases}\infty, & k>0\\ 1, & k=0\\ -\infty, & k<0\end{cases}$$

$$\Rightarrow\lim_{n\to\infty}h\left(\tfrac{1}{n}+k,k\right)\neq\lim_{n\to\infty}h\left(k,\tfrac{1}{n}+k\right)=-\lim_{n\to\infty}\tfrac{1/n^2+2k/n+2k^2}{1/n^2+2k/n}=\begin{cases}-\infty, & k>0\\ -1, & k=0\\ \infty, & k<0\end{cases}$$